

PHASE 3

PROJECT DESIGN PHASE

Components Definition

Gemini Integrations (`gemini_pro.py`, `gemini_flash.py`)

- **Purpose:** To handle the narrative and text generation logic.
- **Responsibility:** Interacting with the Google Gemini API to generate story arcs, dialogue, and image generation prompts.

Image Generator (`image_generator.py`)

- **Purpose:** To synthesize visual panels.
- **Responsibility:** Interfacing with local diffusion models to turn text prompts into comic panel artwork.

Layout Builder (`layout_builder.py`)

- **Purpose:** To construct the visual flow of the comic.
- **Responsibility:** Combining generated text and images into structured pages and panels.

Exporters (**exporters.py**)

- **Purpose:** To format and save the final product.
- **Responsibility:** Converting the built layout into a downloadable PDF format (saved to **static/exports/**).

Web Application (**main.py**, **routes.py**, **schemas.py**)

- **Purpose:** To provide the user interface and handle application data.
- **Responsibility:** Managing HTTP requests/responses, validating data schemas, and serving static assets (CSS).

Module Design

The system is modularized within the **app/** directory, separating configuration (**config.py**), routing (**routes.py**), AI generation (**gemini_pro.py**, **image_generator.py**), and output formatting (**layout_builder.py**, **exporters.py**).

Security Design

- **Environment Variables:** Sensitive keys (like Gemini API keys) are stored securely in a **.env** file, preventing hardcoded credentials in the source code. The **.gitignore** file ensures these are not pushed to version control.